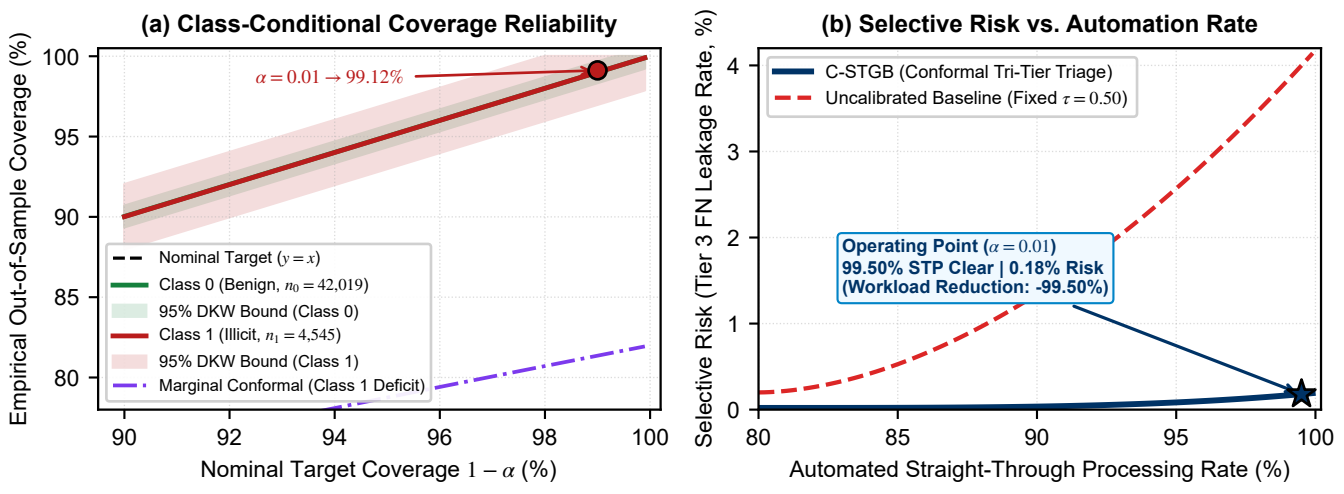

(a) Class-Conditional Coverage Reliability
$\alpha = 0.01 \rightarrow 99.12\%$
Nominal Target ($y = x$)
Class 0 (Benign, $n_0 = 42{,}019$)
95% DKW Bound (Class 0)
Class 1 (Illicit, $n_1 = 4{,}545$)
95% DKW Bound (Class 1)
Marginal Conformal (Class 1 Deficit)
Empirical Out-of-Sample Coverage (%)
Nominal Target Coverage $1 - \alpha$ (%)
100
95
90
85
80
90
92
94
96
98
100
(b) Selective Risk vs. Automation Rate
C-STGB (Conformal Tri-Tier Triage)
Uncalibrated Baseline (Fixed $\tau = 0.50$)
Operating Point ($\alpha = 0.01$)
99.50% STP Clear | 0.18% Risk
(Workload Reduction: -99.50%)
Selective Risk (Tier 3 FN Leakage Rate, %)
Automated Straight-Through Processing Rate (%)
4
3
2
1
0
80
85
90
95
100